

Faculty Workload Allocation, Execution & Academic Effectiveness Management System (FAEMS)- Software

Structured Weekly Workload Allocation and Time Utilization for Faculty Members

Concept

Universities are mandated to ensure **optimal utilization of faculty time** in order to achieve excellence in **teaching, learning, research, mentoring, and academic governance**. However, in the absence of a **clearly defined, structured, and measurable weekly workload framework**, faculty time allocation often becomes uneven, ad hoc, and difficult to monitor. This leads to an overemphasis on teaching alone, while critical academic responsibilities such as **student mentoring, assessment and valuation, research activities, academic preparation, documentation, and administrative contributions** receive inadequate or inconsistent attention.

Additionally, regulatory and accreditation bodies increasingly require **transparent evidence of faculty engagement** across all academic dimensions. Without a standardized weekly timetable aligned to institutional priorities, it becomes challenging to:

- Ensure equitable workload distribution among faculty members
- Protect dedicated time for research and scholarly activities
- Institutionalize student mentoring as a formal academic responsibility
- Systematically plan assessment, evaluation, valuation, and documentation activities
- Demonstrate compliance with quality benchmarks prescribed by bodies such as NAAC, NBA, and UGC

Further, while faculty members are expected to be available on campus for defined working hours, the **lack of a formally approved weekly time-utilization model** results in ambiguity regarding expectations, accountability, and performance assessment.

Therefore, there is a **critical institutional requirement** to develop and implement a **structured, policy-driven weekly workload and time-allocation framework** for every faculty member, based on a **40-hour work week**, with clearly demarcated proportions for:

- Teaching–Learning activities
- Academic preparation, assessment, valuation, and documentation (AAV)
- Student mentoring
- Administrative responsibilities
- Research and scholarly pursuits

Such a framework must be **implementable, auditable, and flexible**, aligned with prescribed working hours (including weekday and Saturday schedules), and capable of serving as a **reference model for faculty timetabling, workload monitoring, performance appraisal, and accreditation reporting**.

1. Weekly Workload Allocation (40 Hours)

Activity Category	% Allocation	Hours / Week
Teaching	40%	16 hours
AAV – Academic Preparation, Assessment, Valuation, Documentation & Data Entry	20%	8 hours
Student Mentoring	15%	6 hours
Administration	10%	4 hours
Research & Scholarly Activities	15%	6 hours
Total	100%	40 hours

2. Working Hours Structure

- **Monday to Friday:** 9:00 AM – 5:00 PM
 - Working hours/day = **7 hours**
 - **Lunch break:** 1:00 PM – 2:00 PM
- **Saturday:** 9:00 AM – 2:00 PM
 - Working hours = **5 hours**

3. Typical Weekly Faculty Timetable (Illustrative) (Standard Work Load)

Monday to Friday (Same Structure, Different Academic Content)

The faculty time table will be different for each faculty member, the work allotted in each slot must be entered based on faculty member time table, however the same number of slots in each category must appear for each member of the faculty.

Time Slot	Monday	Tuesday	Wednesday	Thursday	Friday
9:00 – 10:00	Teaching	Teaching	Teaching	Teaching	Teaching
10:00 – 11:00	Teaching	Teaching	Teaching	Teaching	Teaching
11:00 – 12:00	AAV	Research	AAV	Research	AAV
12:00 – 1:00	Mentoring	AAV	Mentoring	AAV	Mentoring

Time Slot	Monday	Tuesday	Wednesday	Thursday	Friday
1:00 – 2:00	Lunch Break	Lunch Break	Lunch Break	Lunch Break	Lunch Break
2:00 – 3:00	Teaching	Teaching	Teaching	Teaching	Teaching
3:00 – 4:00	Research	Administration	Research	Administration	Research
4:00 – 5:00	AAV	Mentoring	AAV	Mentoring	AAV

Saturday (Focused Academic & Research Day)

Time Slot	Activity
9:00 – 10:00	Research
10:00 – 11:00	Research
11:00 – 12:00	Mentoring / PBL Guidance
12:00 – 1:00	Administration / Academic Meetings
1:00 – 2:00	AAV / Documentation / Planning

4. Weekly Hours Verification (Compliance Check)

Category	Hours Achieved
Teaching	16 hrs
AAV	8 hrs
Mentoring	6 hrs
Administration	4 hrs
Research	6 hrs
Total	40 hrs

✓ Fully compliant with the stated policy

5. Key Implementation Notes (Best Practice)

- **Teaching hours** include lectures, tutorials, labs, and PBL facilitation.
- **AAV time** explicitly covers:

- Lecture preparation
- Question paper setting
- Evaluation & valuation
- LMS/data entry
- Accreditation documentation
- **Mentoring** is scheduled, not incidental (student meetings, career guidance, academic support).
- **Research hours** are protected—no routine meetings should encroach.
- **Saturday** is intentionally lighter and reflective: research, mentoring, planning.

Guiding Principle

Faculty members holding **Admissions, Marketing, Career Development, Student Affairs, or Academic Administrative positions** contribute directly to **institutional growth, student success, and reputation**. Their contributions must be:

- **Formally recognized as workload**
- **Quantified and documented**
- **Balanced against teaching and research**
- **Transparent for appraisal and accreditation**

This requires a **distinct workload category**, not an informal “extra duty”.

Category Definition

Faculty with Major Institutional & Administrative Responsibilities

This category includes faculty serving as:

- Dean / Associate Dean
- Director / Associate Director (School / Program)
- Head of the Department
- Admissions Coordinator / Marketing Lead
- Career Development & Placement Officer
- Student Affairs / Discipline / Welfare Officer

6. Weekly Workload Model – (Administrative & Institutional Role- work load) Faculty

(40 hours per week basis)

Proposed Workload Distribution Table

Activity Category	% Allocation	Hours / Week	Remarks
Teaching	25%	10 hours	Reduced teaching load
AAV – Academic Preparation, Assessment, Valuation & Documentation	15%	6 hours	Course planning, evaluations
Student Mentoring & Career Development	20%	8 hours	Advising, placements, student affairs
Administration & Institutional Roles	25%	10 hours	Dean/HOD/Admissions/Marketing
Research & Scholarly Activities	15%	6 hours	Protected minimum research time
Total	100%	40 hours	

Any changes in the number of hours of work load must be done with the approval of the higher authorities based on the justifiable request made by the member of the Faculty.

7. Faculty in Full-Time Institutional Roles (Full Time Role-work load)

Category	%
Administration & Institutional Roles	50–60%
Teaching	15–20%
Mentoring & Career Development	10–15%
Research	10–15%

Used for:

- Full-time Deans
- Admissions Heads during admission cycle
- Placement Directors during peak seasons

Software Proposal:

Proposed Software Blueprint

Faculty Workload Allocation, Execution & Effectiveness Management System (FW-EEMS)

1. Purpose of the Software

The system is intended to:

- **Allocate faculty workload** based on institutional policies (40-hour model, differentiated roles)
 - **Ensure execution** of allotted work (not just allocation)
 - **Measure effective utilization of time**
 - **Derive faculty effectiveness metrics** (not simplistic “efficiency”)
 - **Provide evidence for governance, appraisal, and accreditation**
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2. Appropriate Terminology (Very Important)

Instead of using only *efficiency* (which implies speed/output), the following terms are **academically appropriate**:

Recommended Core Metric Name

👉 **Faculty Academic Effectiveness Index (FAEI)**

Supporting Metrics

- **Time Utilization Index (TUI)**
- **Workload Fulfilment Ratio (WFR)**
- **Academic Contribution Score (ACS)**
- **Role Responsibility Fulfilment (RRF)**

These avoid the perception of faculty being treated as factory labor and are more acceptable in academic ecosystems.

3. Major Software Modules

Module 1: Faculty Profile & Role Management

Purpose: Define who the faculty member is and what roles they hold.

Key Data Elements

- Faculty ID, designation, department

- Teaching role (UG/PG/PhD)
- Administrative roles (HoD, Dean, Admissions, CDC, etc.)
- Role tenure (start–end dates)
- Applicable workload model (Standard / Administrative / Full-time Role)

✦ *This module determines which workload distribution table applies.*

Module 2: Workload Policy & Rule Engine

Purpose: Encode institutional workload rules into the system.

Capabilities

- Define workload categories:
 - Teaching
 - AAV
 - Mentoring
 - Administration
 - Research
- Assign percentage & hour limits
- Role-based overrides (Dean, HoD, Admissions Lead)
- Semester-wise or annual policies

✦ *This acts as the brain of the system.*

Module 3: Automated Workload Allocation Engine

Purpose: Convert policy into **individual weekly faculty workload plans**.

Inputs

- Faculty profile
- Timetable constraints
- Academic calendar
- Role responsibilities

Outputs

- Weekly workload breakup (hours & activities)
- Individual faculty timetable (draft)
- Department-level workload summary

✦ *Manual changes allowed only with approval and audit trail.*

Module 4: Execution & Activity Logging Module

Purpose: Ensure **work done = work logged**, without micromanagement.

How it Works

Faculty log activities under predefined categories:

- Teaching sessions (auto-fetched from timetable)
- Assessment / valuation tasks
- Mentoring sessions (student-wise or batch-wise)
- Research activities (paper writing, proposal, experiments)
- Administrative actions (meetings, admissions events, reports)

Features

- Simple daily/weekly logging
- Evidence attachment (optional)
- Mobile & desktop friendly

✦ *This is NOT attendance; it is academic activity logging.*

Module 5: Time Utilization & Fulfilment Analytics

Purpose: Compare **planned vs executed** workload.

Key Calculations

- Planned Hours vs Actual Logged Hours
- Category-wise fulfilment %
- Role-wise deviation analysis
- Peak-load detection (overburdened faculty)

Outputs

- Time Utilization Index (TUI)
 - Workload Fulfilment Ratio (WFR)
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Module 6: Academic Effectiveness & Contribution Engine

Purpose: Translate time + outcomes into **meaningful academic indicators**.

Sample Inputs

- Teaching delivery completion
- Student feedback (normalized)
- Assessment timeliness
- Mentoring outcomes
- Research outputs (not raw counts)
- Administrative deliverables

Key Output

Faculty Academic Effectiveness Index (FAEI)

A weighted composite score, customizable by institution.

✦ *This avoids reducing faculty to “hours only”.*

Module 7: Dashboards & Decision Support

Purpose: Role-based visibility without overexposure.

Dashboards

- Faculty (self-view)
- HoD
- Dean / Director
- Registrar / VC

Insights

- Under-utilization / over-utilization
 - Faculty load imbalance
 - Departmental stress points
 - Evidence for appraisal & promotions
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Module 8: Compliance, Accreditation & Reporting

Purpose: One-click evidence generation.

Reports

- NAAC Criterion-wise faculty engagement
- NBA workload compliance
- Faculty contribution statements
- Departmental workload audit reports

✦ *This module alone saves hundreds of man-hours.*

4. High-Level Software Flow (Textual)

Faculty Profile + Role Assignment



Workload Policy Engine



Automated Workload Allocation



Faculty Weekly Plan & Timetable



Activity Execution & Logging



Time Utilization & Fulfilment Analysis



Academic Effectiveness Index Calculation



Dashboards, Appraisal & Accreditation Reports

5. Software Architecture (Conceptual)

Architecture Type

👉 **Modular, Role-Based, Rule-Driven Web Application**

Suggested Layers

- **Presentation Layer** – Web & Mobile UI
 - **Business Logic Layer** – Workload rules, analytics, scoring
 - **Data Layer** – Faculty, workload, logs, evidence
 - **Integration Layer** – LMS, ERP, HRMS (optional)
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6. Governance & Safeguards (Critical)

- No minute-level tracking (academic dignity preserved)
- Focus on **outcomes + accountability**

- Role-based access control
 - Full audit trail
 - Policy changes versioned
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7. Suggested Software Name Options

- **FAEMS** – Faculty Academic Effectiveness Management System
- **F-WISE** – Faculty Workload & Institutional Support Engine
- **ACADEMETRIX** – Academic Workload & Effectiveness Platform
- **PROFAC** – Professional Faculty Contribution System

3. Software Requirement Specification (SRS)

Faculty Workload Allocation, Execution & Academic Effectiveness Management System (FAEMS)

1. Introduction

1.1 Purpose

The purpose of this software is to **systematically allocate, monitor, and evaluate faculty workload** in alignment with institutional policies, ensuring:

- Fair and transparent workload allotment
- Effective execution of assigned academic and administrative responsibilities
- Measurable utilization of faculty time
- Evidence-based evaluation of faculty academic effectiveness

The system shall support **teaching, research, mentoring, administration, admissions, marketing, student affairs, and career development activities.**

1.2 Scope

The system shall:

- Operate on a **40-hour weekly workload framework**
 - Support **multiple workload models** (regular faculty, HoD, Dean, administrative roles)
 - Capture **planned vs executed work**
 - Compute **academic effectiveness indices**
 - Generate reports for **governance, appraisal, NAAC/NBA accreditation**
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1.3 Definitions (Key Terms)

Term	Definition
FAEI	Faculty Academic Effectiveness Index (composite indicator)
TUI	Time Utilization Index
WFR	Workload Fulfilment Ratio
AAV	Academic preparation, Assessment, Valuation, Documentation

Term	Definition
Role-based workload	Workload distribution based on institutional responsibility

2. User Roles

Role	Access Level
Faculty Member	View plan, log work, self-dashboard
HoD	Department view, approvals, monitoring
Dean / Director	School-level oversight
Registrar / VC	Institutional analytics
Admin (System)	Configuration & policy management

3. Functional Requirements

3.1 Faculty Profile & Role Management

- Create and manage faculty profiles
 - Assign multiple roles (teaching + administrative)
 - Define role tenure (start/end dates)
 - Map faculty to workload model automatically
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3.2 Workload Policy & Rule Engine

- Define workload categories and percentages
 - Support multiple workload templates
 - Allow policy versioning (semester/year-wise)
 - Prevent policy violation (hard constraints)
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3.3 Automated Workload Allocation

- Generate weekly workload plans per faculty
- Allocate hours across categories
- Generate draft faculty timetable

- Allow role-based teaching load reduction
 - Approval workflow (HoD/Dean)
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3.4 Activity Execution & Logging

- Faculty to log activities under predefined categories
 - Teaching auto-validated via timetable
 - Manual logging for research, mentoring, admin
 - Evidence attachment (optional)
 - Weekly submission with lock-in
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3.5 Time Utilization & Fulfilment Analytics

- Compare planned vs executed hours
 - Category-wise fulfilment
 - Identify under/over-utilization
 - Generate TUI and WFR automatically
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3.6 Academic Effectiveness Engine

- Compute **Faculty Academic Effectiveness Index (FAEI)**
 - Weight-based scoring (configurable)
 - Include qualitative + quantitative inputs
 - Avoid raw hour-based ranking
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3.7 Dashboards & Reports

- Faculty self-dashboard
 - Department workload heatmaps
 - School/institution analytics
 - NAAC/NBA-ready reports
 - Export to PDF/Excel
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4. Non-Functional Requirements

4.1 Usability

- Simple UI (academic-friendly)
- Web + mobile responsive
- Minimal daily input (<5 minutes)

4.2 Performance

- Handle institutional-scale data
- Response time <3 seconds

4.3 Security

- Role-based access control
- Audit trail for all changes
- Secure authentication

4.4 Compliance

- Align with NAAC, NBA, NEP 2020
- Preserve academic dignity (no micromanagement)

5. Data Requirements (High-Level)

Core Entities

- Faculty
- Role
- Workload Policy
- Weekly Plan
- Activity Log
- Effectiveness Metrics
- Reports

6. Academic Effectiveness Metrics (Indicative)

Metric	Description
TUI	Actual hours ÷ Planned hours
WFR	Tasks completed ÷ Tasks assigned
ACS	Academic Contribution Score
RRF	Role Responsibility Fulfilment

Metric	Description
FAEI	Weighted composite index

7. System Constraints & Safeguards

- No minute-by-minute tracking
 - No attendance-only logic
 - Outcome-weighted evaluation
 - Transparent faculty access to own data
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8. Future Enhancements (Optional)

- LMS integration
 - HRMS linkage
 - AI-based workload prediction
 - Faculty burnout alerts
 - Research outcome mapping
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9. Deliverables from Vendor / Developer

- Fully functional web application
- Admin configuration panel
- User manuals
- Training & deployment support
- Source code & documentation

Faculty Academic Effectiveness Index (FAEI) (Measurement)

1. Philosophy Behind FAEI (Very Important)

FAEI is **not a productivity score** and **not a time-only metric**.
It measures:

How effectively a faculty member converts allotted time and roles into meaningful academic, institutional, and student outcomes.

Hence, it combines:

- **Planned vs executed effort**
- **Quality of academic contribution**
- **Role responsibility fulfilment**
- **Outcomes, not just activity**

2. High-Level FAEI Formula

$$FAEI = w_1(TUI) + w_2(WFR) + w_3(ACS) + w_4(RRF)$$

Where:

Symbol	Meaning
TUI	Time Utilization Index
WFR	Workload Fulfilment Ratio
ACS	Academic Contribution Score
RRF	Role Responsibility Fulfilment
$w_1 + w_2 + w_3 + w_4 = 1$	Total weight = 100%

3. Recommended Weightage Structure (Balanced & Safe)

Standard Faculty (Teaching + Research + Mentoring)

Component	Weight (%)
TUI	20%
WFR	25%
ACS	40%
RRF	15%
Total	100%

Why this works:

Outcomes (ACS) dominate, time supports—not controls—evaluation.

Faculty with Administrative / Institutional Roles

Component	Weight (%)
TUI	20%
WFR	20%
ACS	30%
RRF	30%
Total	100%

Administrative accountability is explicitly recognized.

4. Component Definitions & Sub-Formulas

4.1 Time Utilization Index (TUI)

$$\text{TUI} = \frac{\text{Actual Logged Hours}}{\text{Planned Hours}}$$

Rules

- Capped at **1.10** (no reward for unhealthy overload)
 - Below **0.80** triggers advisory, not penalty
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4.2 Workload Fulfilment Ratio (WFR)

$$WFR = \frac{\text{Tasks Completed}}{\text{Tasks Assigned}}$$

Tasks include

- Courses delivered
- Evaluations completed
- Mentoring sessions held
- Assigned admin deliverables

4.3 Academic Contribution Score (ACS)

(Most important component)

$$ACS = 0.35(T) + 0.25(R) + 0.20(M) + 0.20(AAV)$$

Where:

Symbol	Contribution Area
T	Teaching effectiveness
R	Research & innovation
M	Mentoring & student development
AAV	Academic preparation & assessment quality

Each sub-score is **normalized between 0 and 1**.

Teaching Effectiveness (T)

- Course delivery completion
- Student feedback (normalized, bias-controlled)
- Learning outcome attainment

Research & Innovation (R)

- Quality-weighted publications
- Proposals, patents, supervision
- Progress-based, not count-based

Mentoring (M)

- Number and quality of mentoring interactions
- Student progression indicators

AAV Quality

- Timeliness of evaluation
 - Quality of question papers
 - Documentation completeness
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4.4 Role Responsibility Fulfilment (RRF)

$$\text{RRF} = \frac{\text{Role Deliverables Achieved}}{\text{Role Deliverables Assigned}}$$

Applies strongly to:

- HoD
 - Dean
 - Admissions / Marketing leads
 - Career development officers
 - Student affairs roles
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5. Final FAEI Score Interpretation

FAEI Range	Interpretation
≥ 0.85	Highly Effective
0.70 – 0.84	Effective
0.55 – 0.69	Satisfactory
< 0.55	Needs Support (Not Failure)

Policy Rule:

FAEI **must not be used alone** for punishment. It is a **developmental index**.

6. Why This Model Is Defensible

- ✓ Academically respectful
- ✓ Avoids micromanagement

- ✓ Outcome-weighted
 - ✓ Role-sensitive
 - ✓ Accreditation-friendly
 - ✓ Software-implementable
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7. How This Fits into Software

- Each component is **modular**
 - Weightages are **configurable**
 - Role-based switching is automatic
 - Year-on-year trends can be generated
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8. Recommended Governance Clause (Optional)

“FAEI shall be used as a formative academic effectiveness indicator for faculty development, planning, and institutional quality assurance, and shall not be used as a standalone punitive metric.”